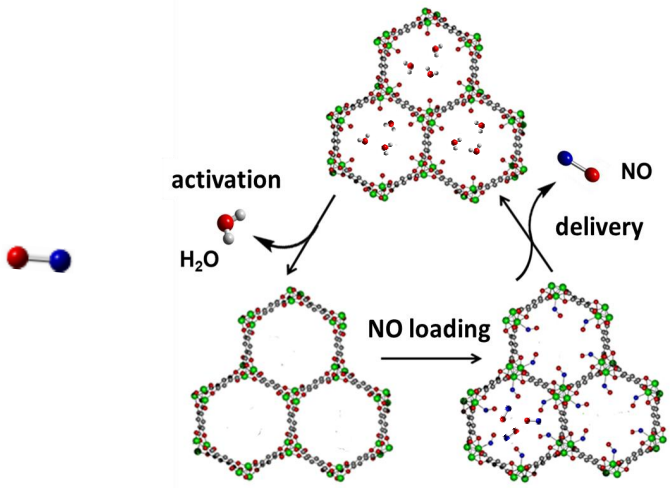
PolyMOF 2: Q8 Updates Modelling

Simulating Nitric Oxide diffusion within MOF-based Polymers Used to Prevent Catheter Related Thrombosis and Infection



Presenter: Nisarg Makwana

Supervisors: Professor G Menary

Dr A Lennon

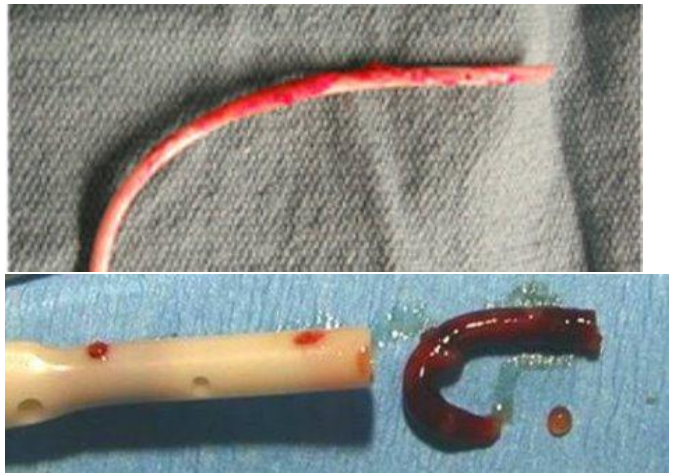
Queens University Belfast: School of Mechanical & Aerospace Engineering

Polymer Processing Research Centre, Queen's University, Belfast, UK

School of Chemistry, University of St Andrews, Scotland, UK

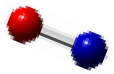
Overview : Antithrombosis Effect of NO

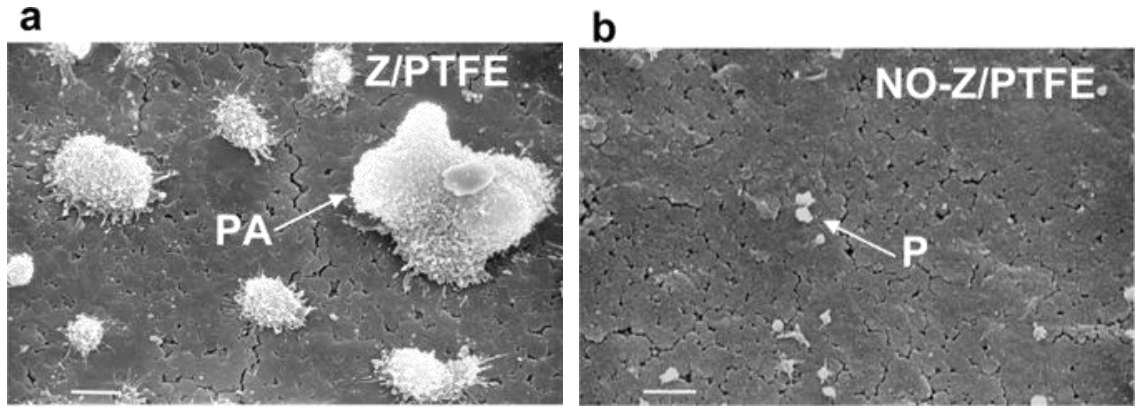
Thrombus & Infection during to Catheterization



- Formation of biofilm on Catheter surfaces
- Long term contact with Catheter
- Coarse Catheter surface
- Shear force while catheter insertion in Bloodstream

Introduction of Nitric Oxide

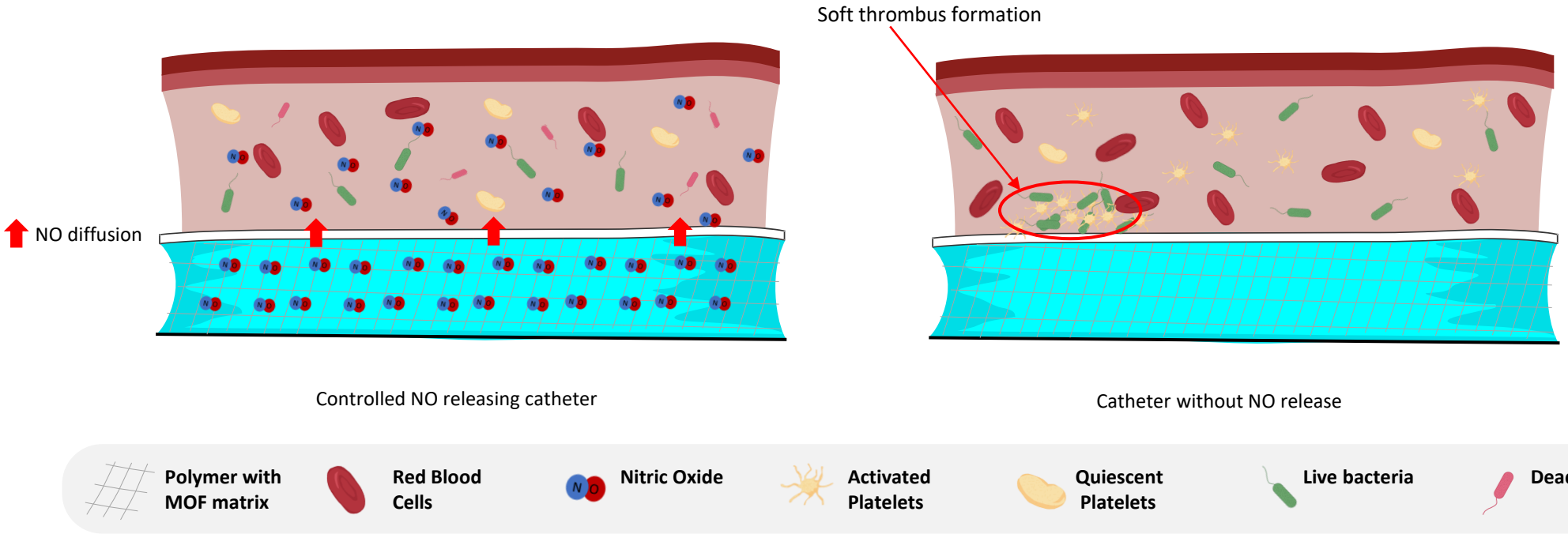




Scanning electron micrographs of the surface of (a) untreated-zeolite-A/PTFE disks (Z/PTFE), (b) NO-loaded Co-zeolite-A/PTFE. [1]

- Stops Coagulation of Inflammatory cells
- Potent inhibitor of Platelet Activation and Adhesion
- Natural Antimicrobial agent in the Immune system
- Kills live Bacteria

Overview : Antithrombosis Effect of NO

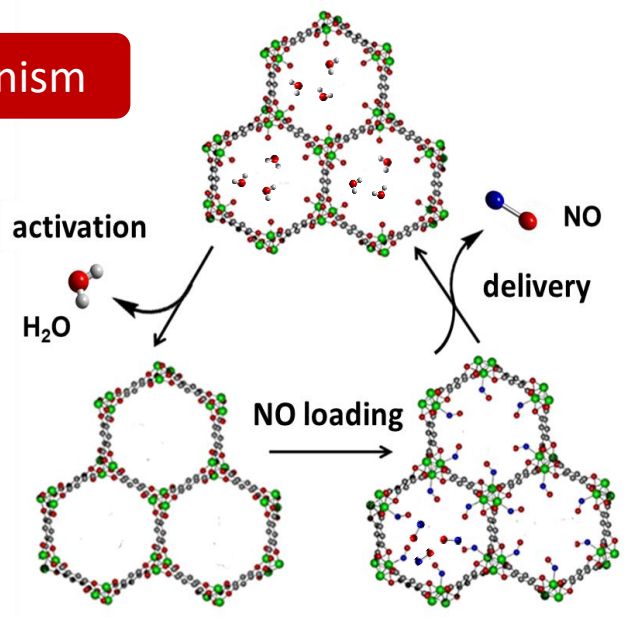


- Activated platelets aggregation causes thrombus formation
- Nitric Oxide in low levels inhibits growth of bacteria and platelets activation
- Nitric oxide prevents activation of platelets, that ultimately leads to thrombus formation
- Excess Nitric Oxide Concentration is Toxic

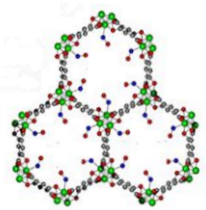
Overview : MOF(Metal Organic Frameworks) manufacturing and NO loading

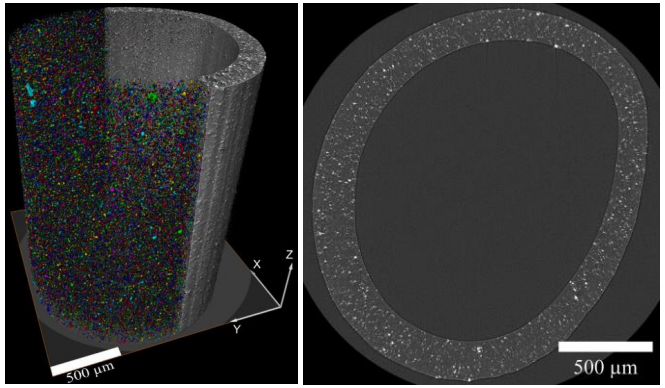
NO Delivery Mechanism

- Loading of NO into MOF
- Delivery of NO into target
- Activation of MOF via H2O removal



MOF micro-CT





Manufacturing

Tube Dimensions:1.75 mm OD with 1.25mm ID

1,2,5 % wt. MOF-polymer blending

Introduction : Project Summary

QUEEN'S UNIVERSITY BELFAST

poly M2F

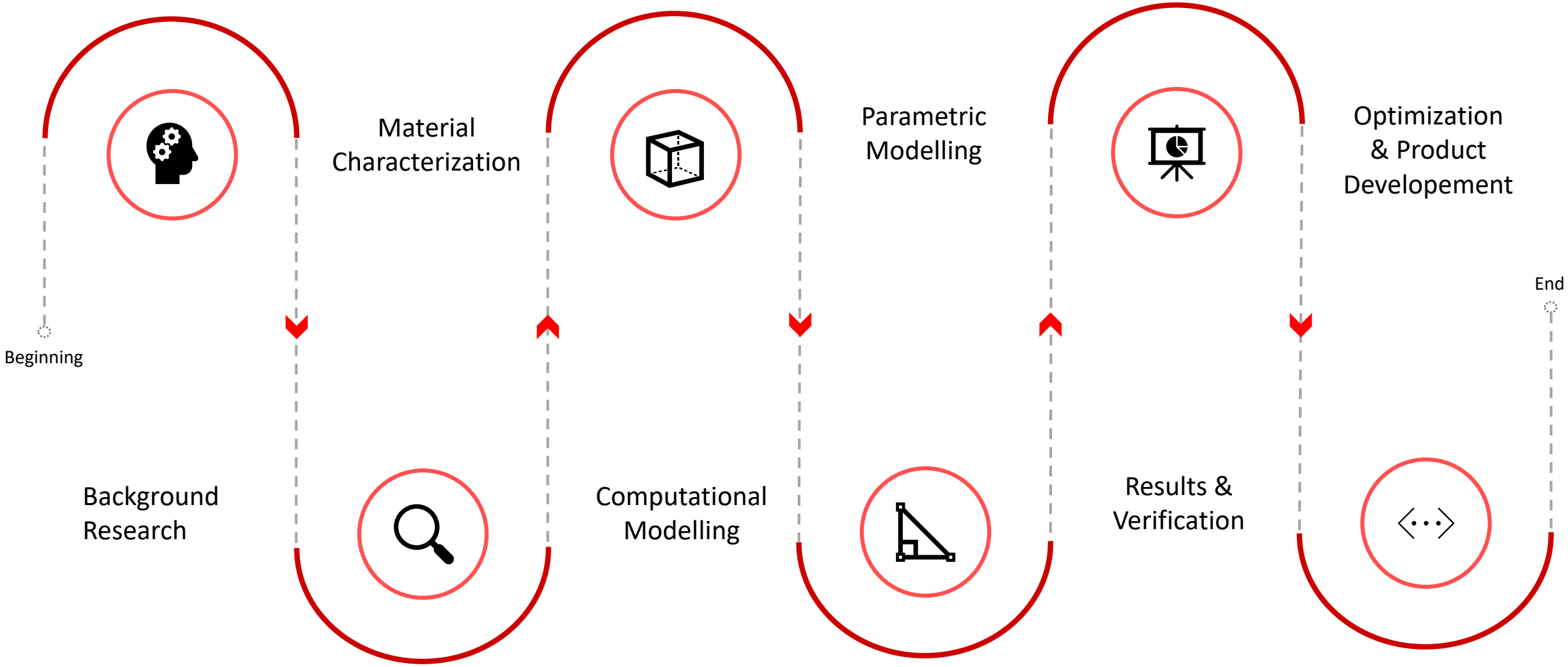
AIM

The project will develop a novel computational model to mitigate problems for catheter related Thrombosis and Infection by Surrogate model, experimental verification and validation for MOF based polymer as material substitute in a Catheter.

OBJECTIVES

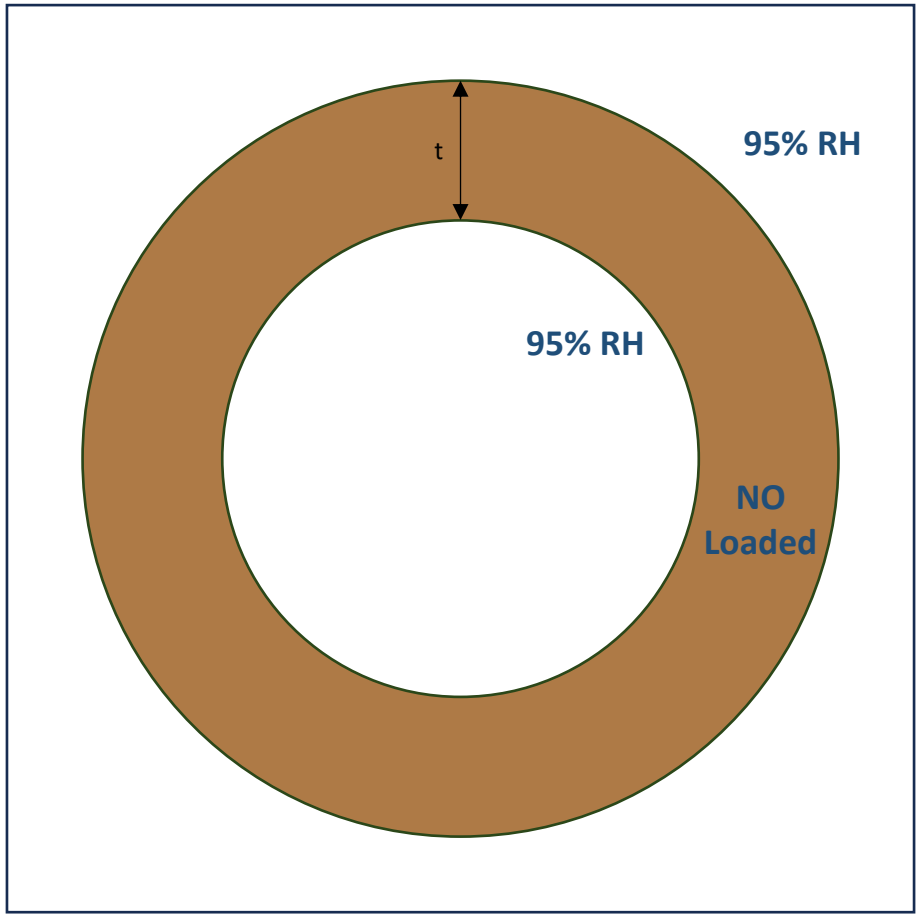
1. Multiphase spatio-temporal modelling to predict and tune NO release.
2. Prediction of optimised MOF loading polymer matrix combination to give a desired NO release.
3. NO and water vapour transmission experiments and computational modelling of NO release with verification.
4. Pre-prototype development for implementation in Medical Device.

Introduction : Modelling Roadmap

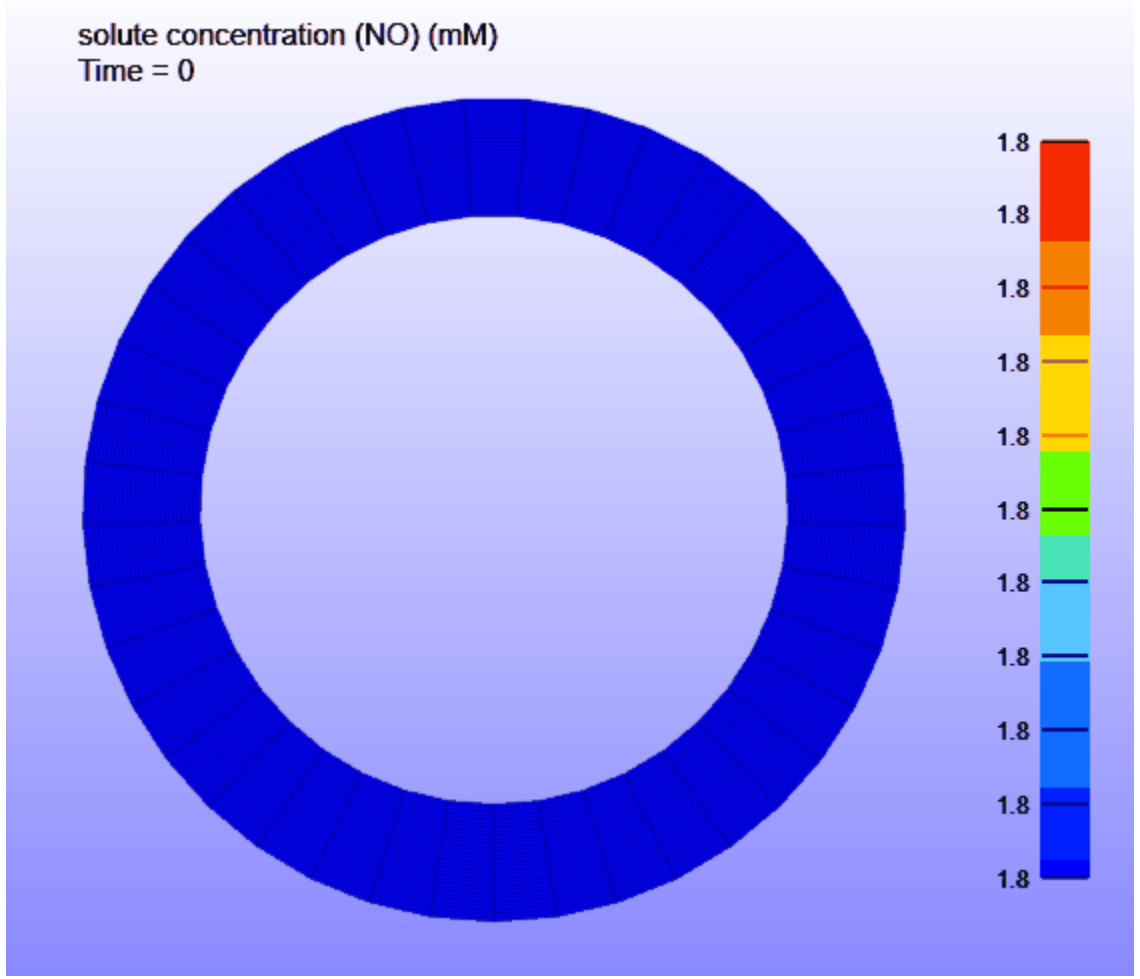


Model Construction: Base Model Modelling Physiosorbed NO

Pebax Tube

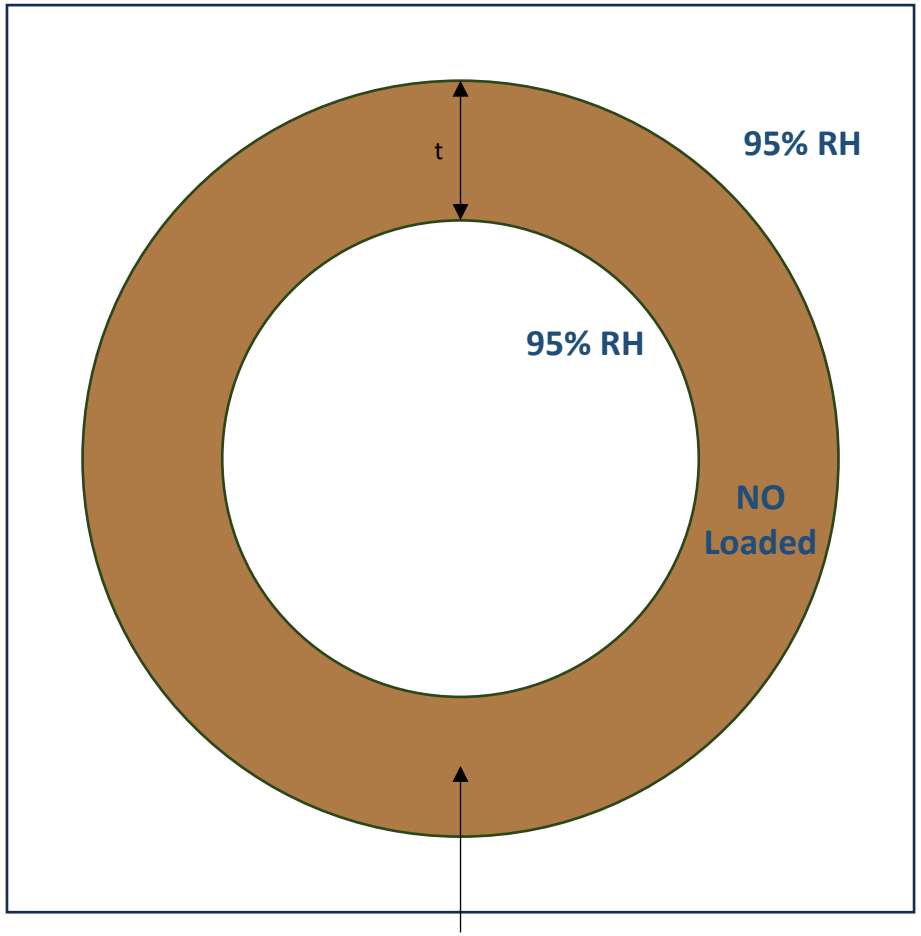


Diffusible NO Concentration



Model Construction: Base Model Modelling Chemisorbed NO

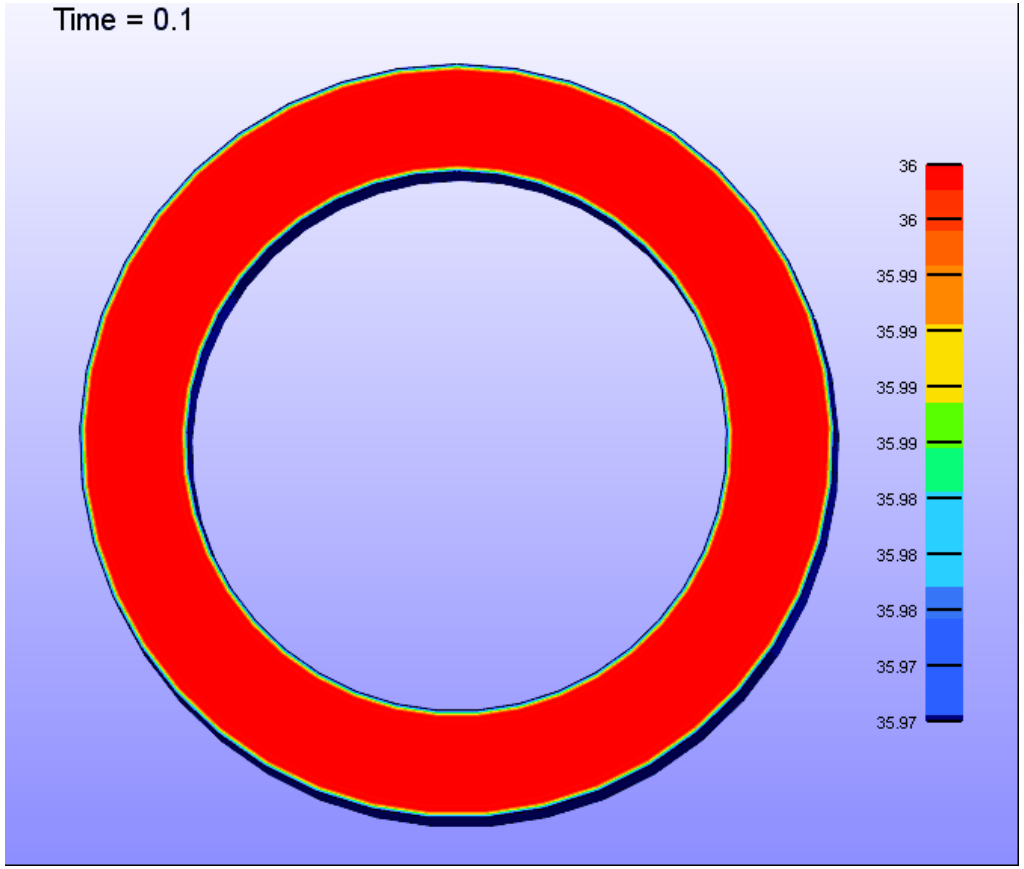
Pebax Tube



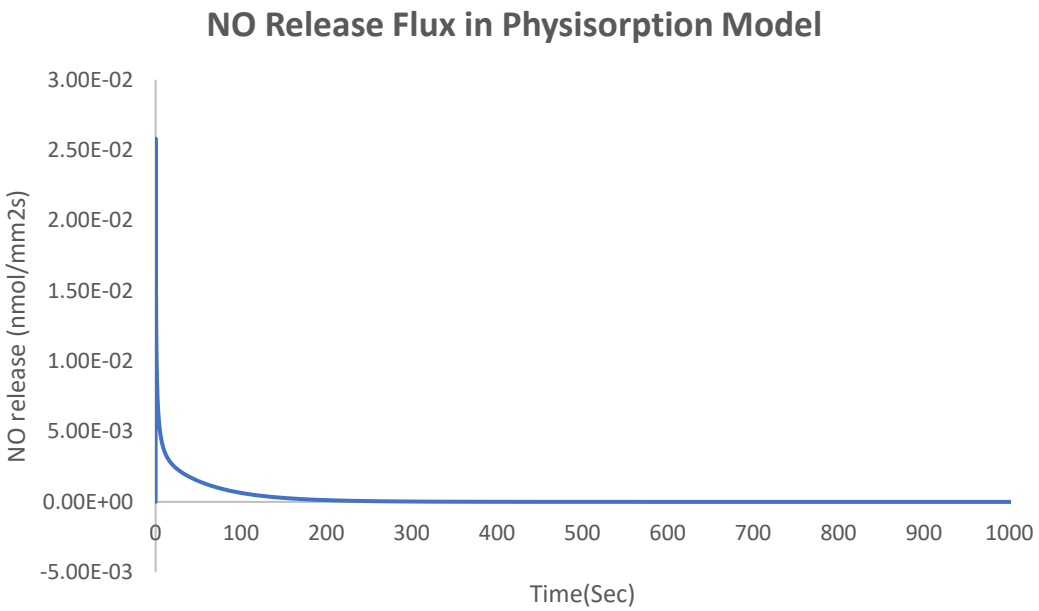
Reaction defined in the material

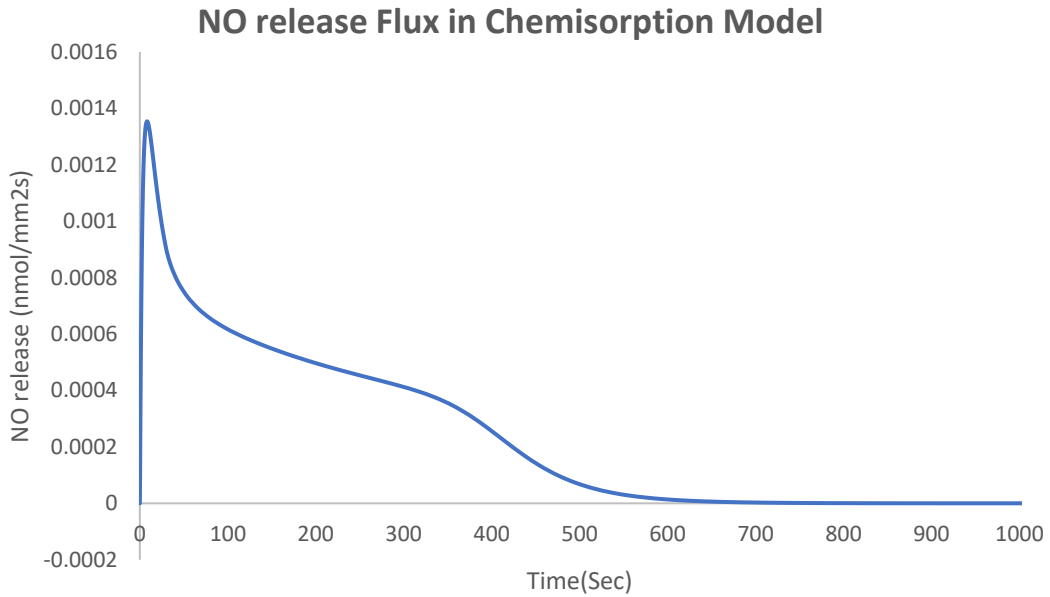
(Material + NO_I)+ H2O → (Material + H2O) + NO_R

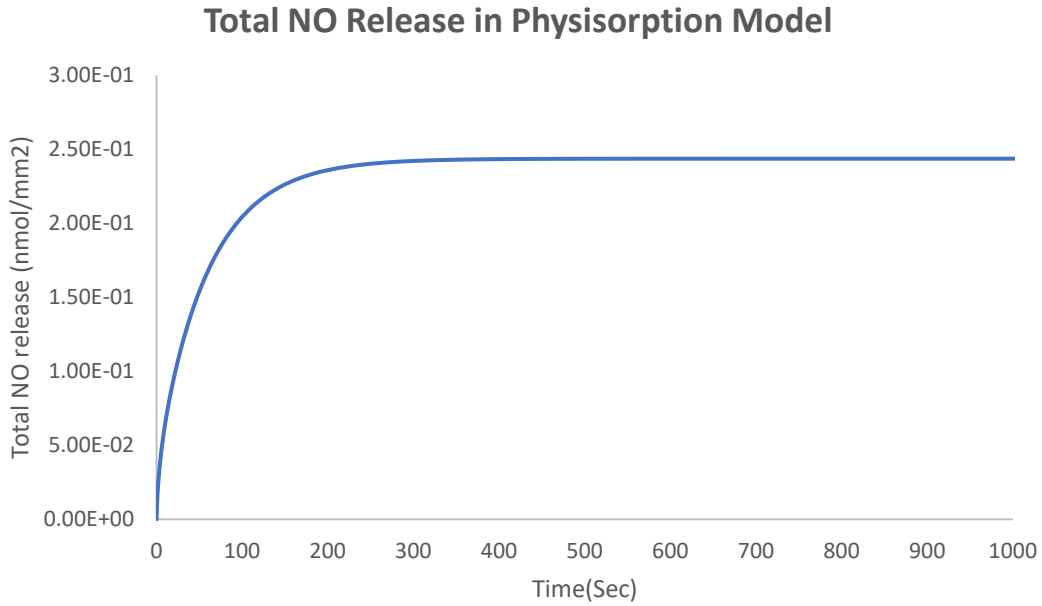
Diffusible NO Concentration

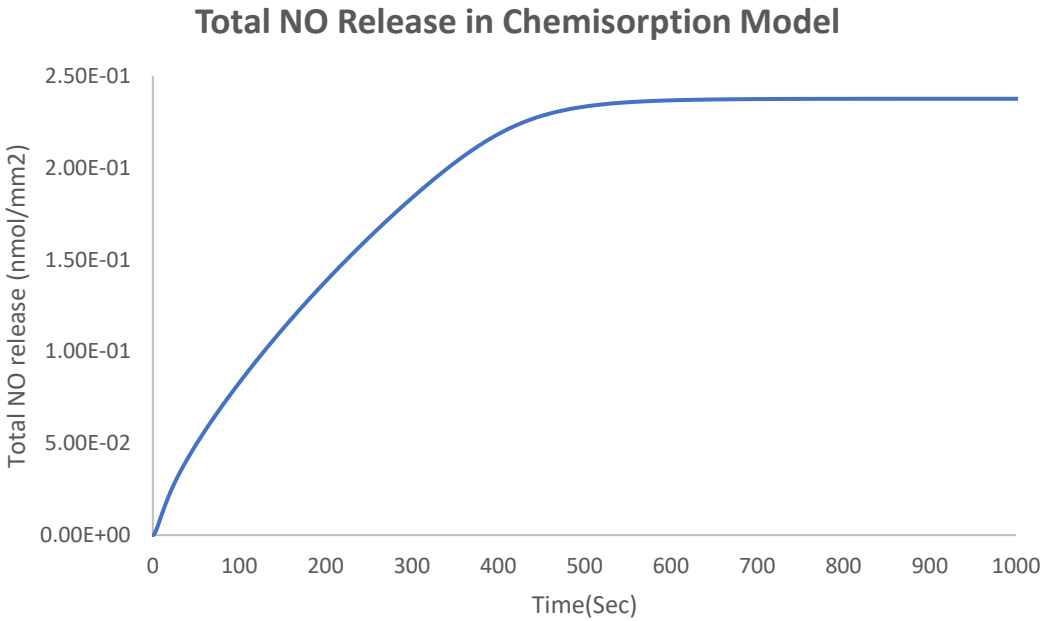


Model Construction: Comparing both Model





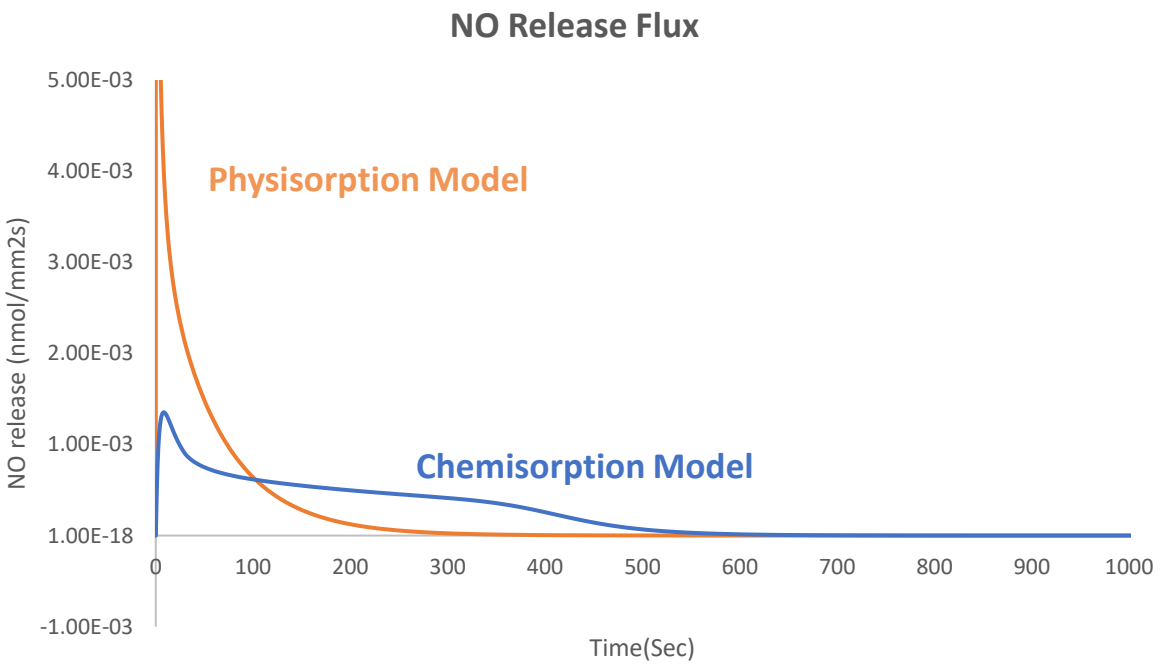


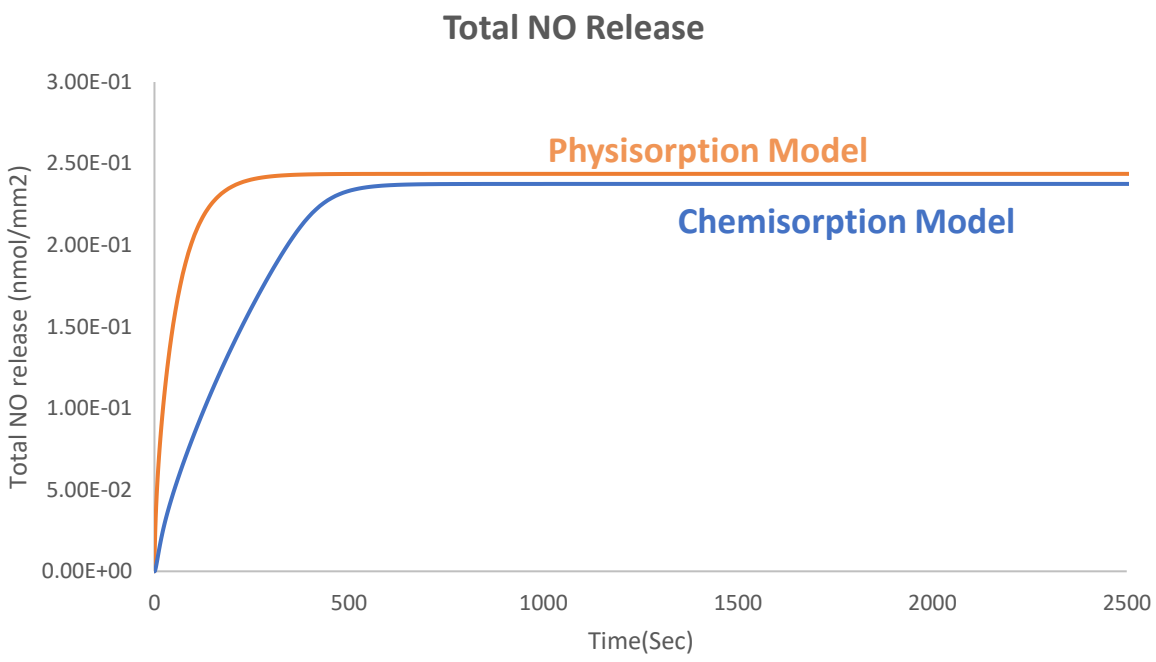


Model Construction: Comparing both Model

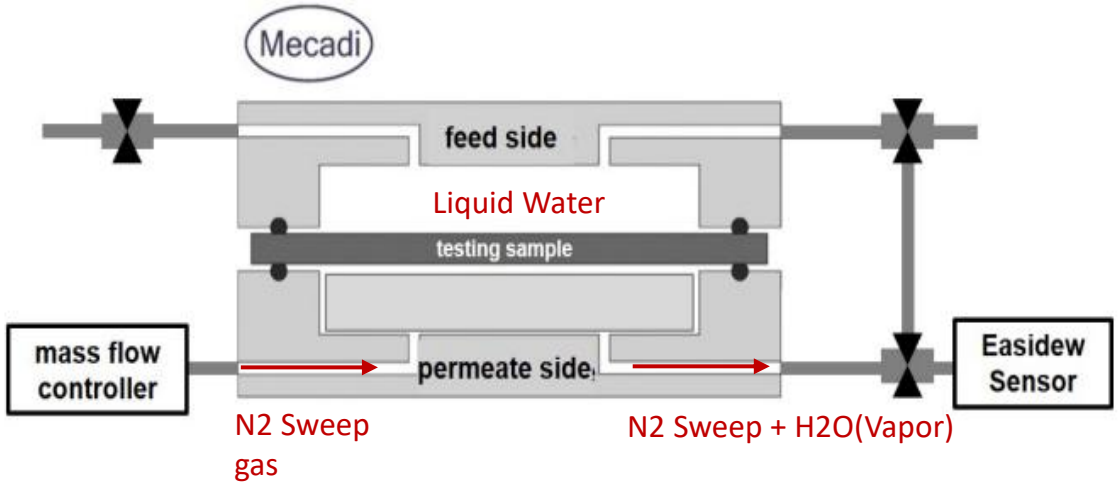
Physisorption Model: The NO Releases spikes up as soon as the Model is run, a much quicker release pattern is seen because of higher diffusion coefficient of NO in the blank polymer matrix

Chemisorption Model: The NO Release lasts for longer time because of H2O's Involvement. The NO release pattern is highly driven by the H2O Diffusion Coefficient



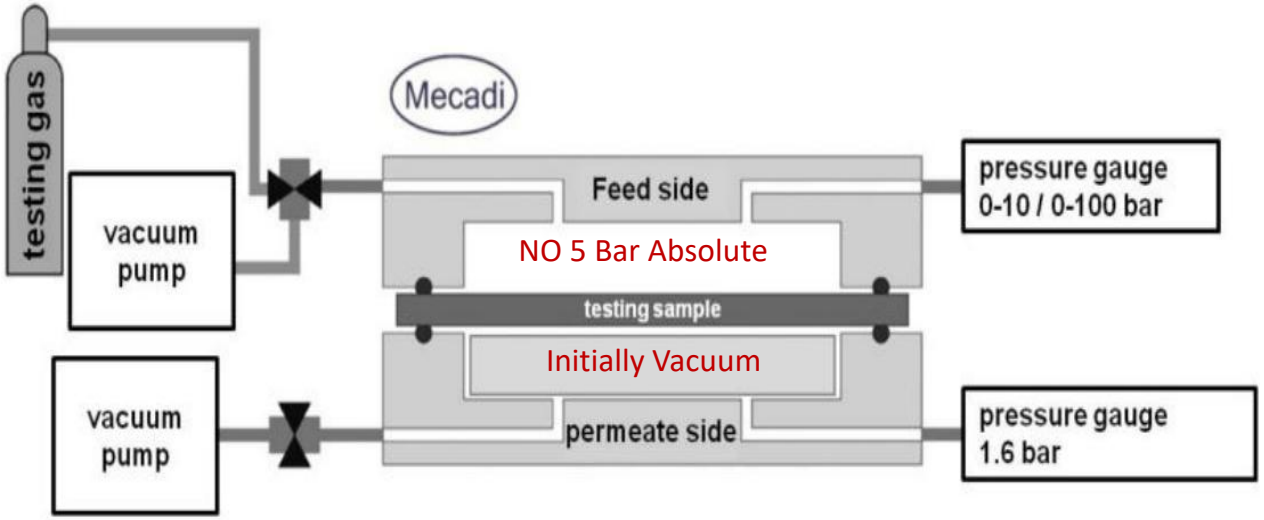


Experimental testing : Diffusion Tests



H2O Diffusion Tests

- ISO 15106-1 Standard is used in determination of the water vapour permeability of the sample material.
- H2O measurement are carried out with Michell Easidew dew point transmitter for trace concentrations (dew point range 20 ◦C to -100 ◦C).

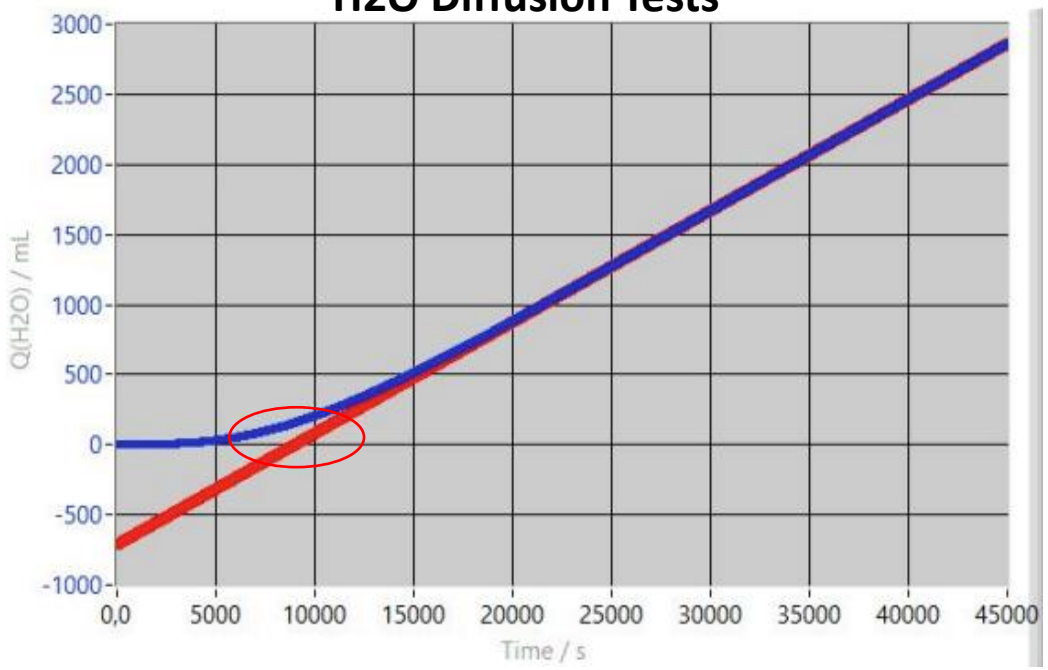


NO Diffusion Tests

- DIN 53380/ISO 15105(Pressure increase) Standard is used in determination of the NO permeability of the sample material.
- NO measurement are carried out with Pressure sensor , Pressure Transducer permeate side: WIKA D-10-P; 0-1.6 bar absolute.

Experimental testing : Diffusion Tests

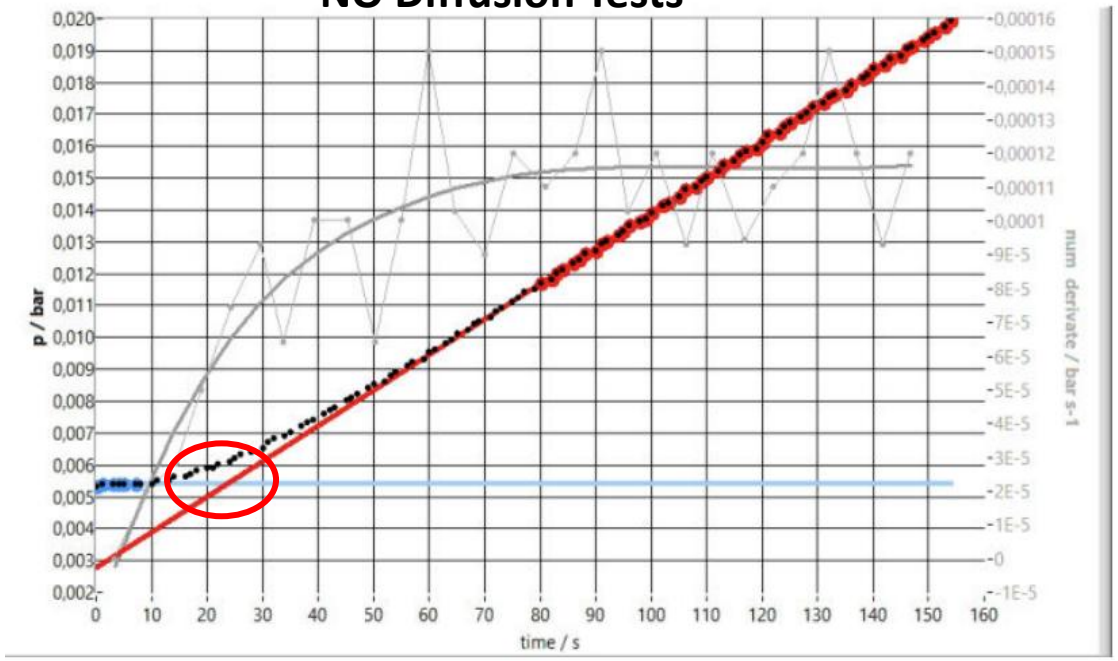
H2O Diffusion Tests



Graph 1. Represents the Numerical Integral of Water flux across the test sample over time.

Sample Id: 2181-240620-004

NO Diffusion Tests



Graph 2. Represents the Pressure within the permeate side over time.

Sample Id: 2062-240925-001

Diffusion Coefficient of Material

$$D(P) = \frac{d^2}{6 \cdot \Theta}$$

The breakthrough time Θ is determined by linearly extrapolating the steady (quasi-stationary) portion of the curve to intersect the time axis. It represents the time at which significant permeation starts.

Effective Solubility Coefficient of Material

$$S = \frac{\text{Dissolved solute concentration within the material}}{\text{Concentration of solute on the surface of the Material}}$$

NO release for blank Pebax 5533 95% RH

Diffusivity of Material

NO Diffusivity : 1.2e-6 cm2/s
From Mecadi Tests

Solubility of Material

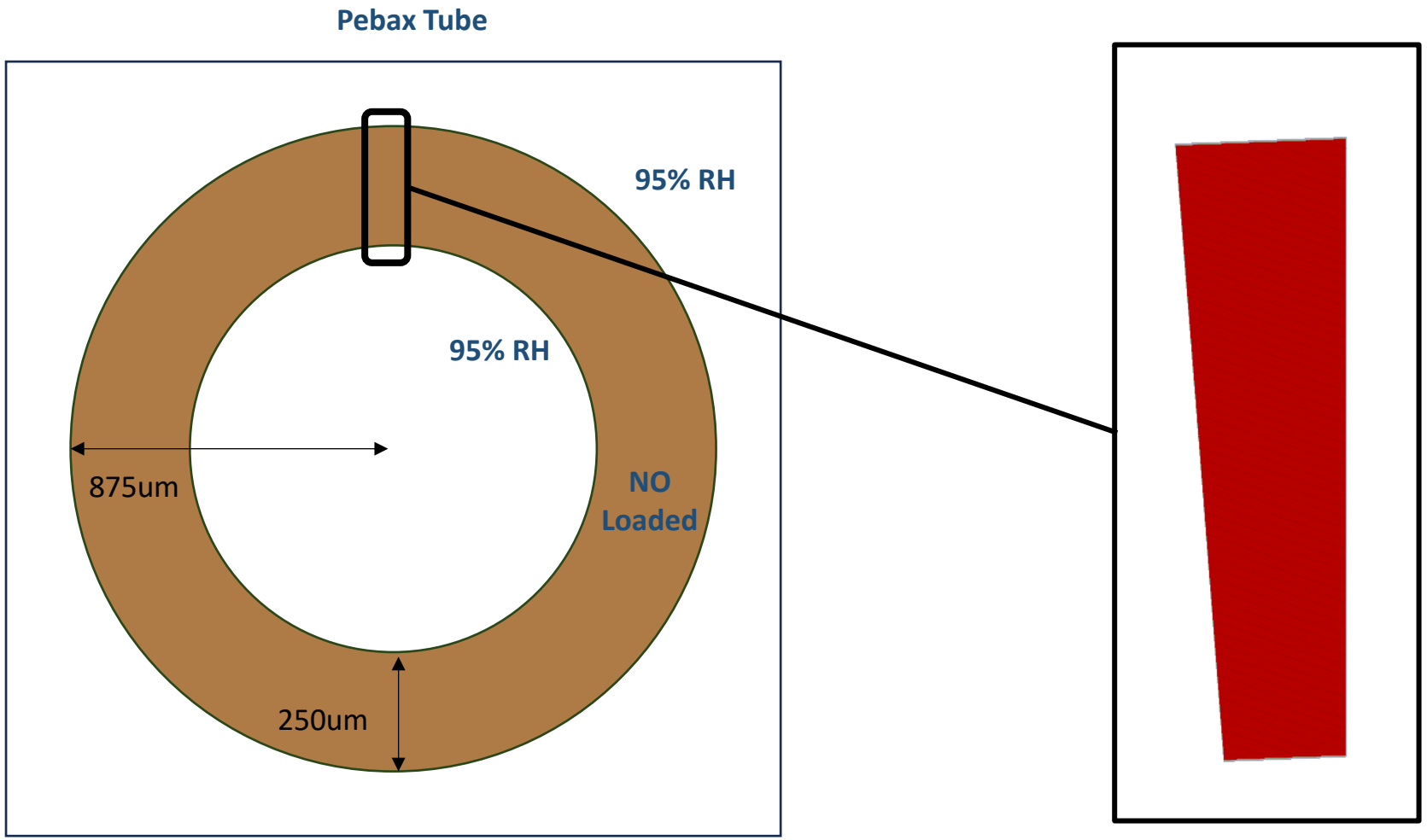
NO Solubility : Derived From
Mecadi Results

NO Loading / IC

Total NO release from
Experiments is Input for
the Model:

Release Conditions/BC

95% RH



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Solubility of Material

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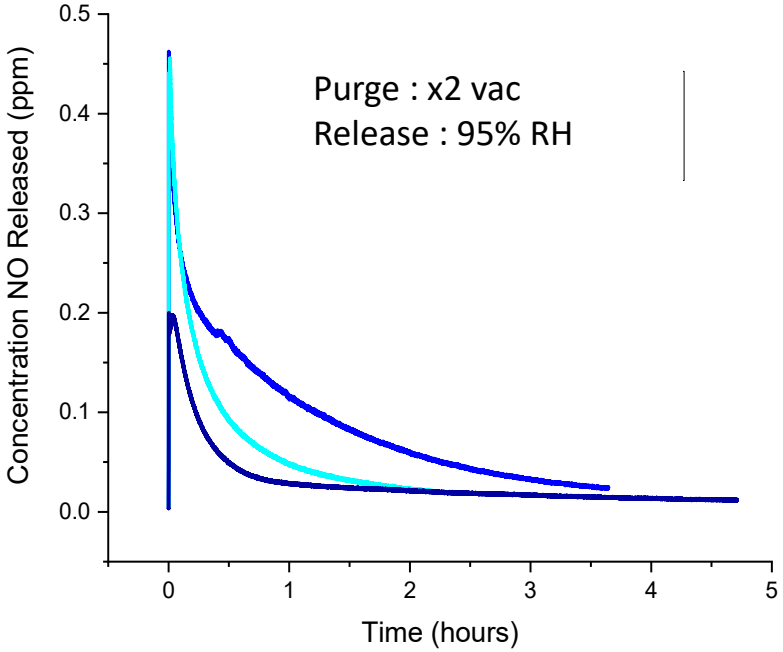
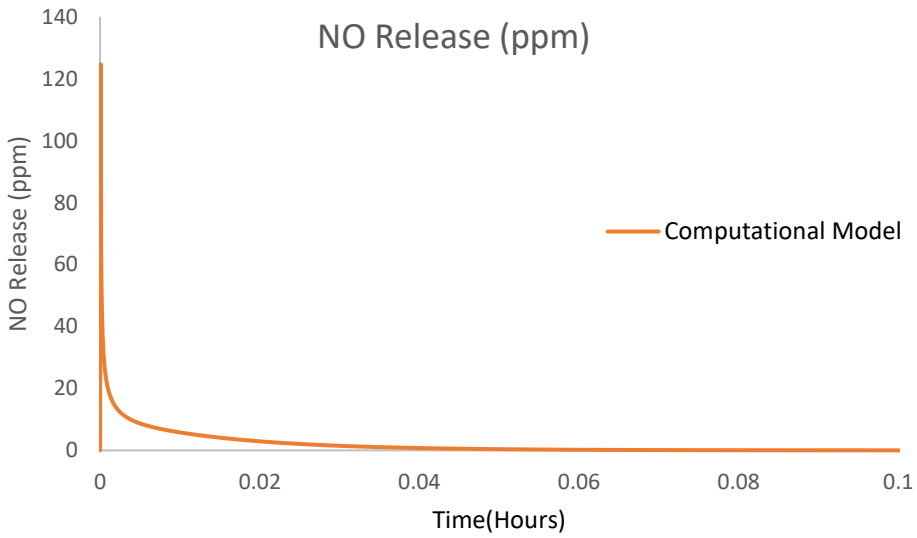
Total NO release from Experiments is Input for the Model:

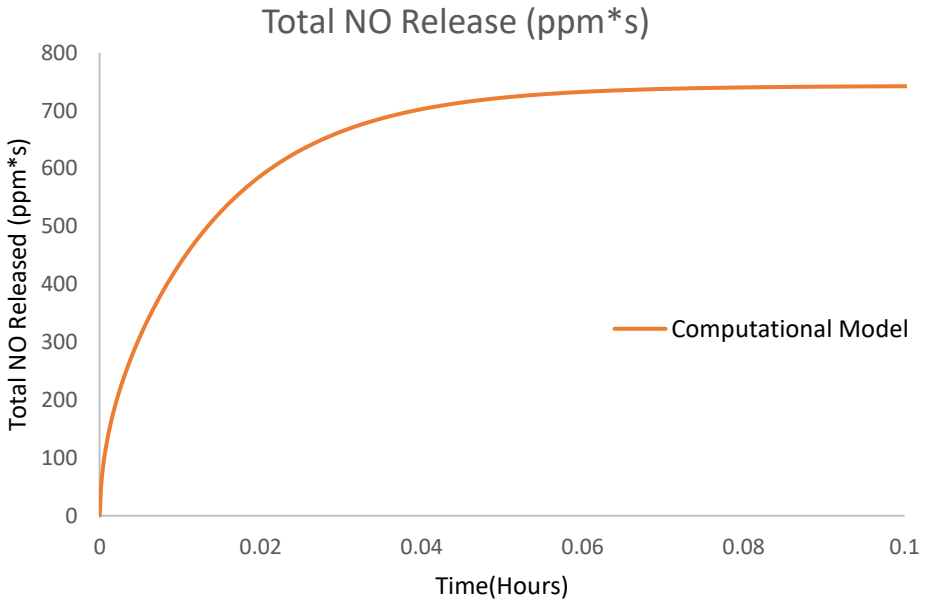
Release Conditions/BC

95% RH

Dehydration: 80°C, 16 hr, x10^{-3} torr
NO Loading : 1 hr, 2 bar, 100%







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From Mecadi Tests

Solubility of Material

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NO Loading / IC

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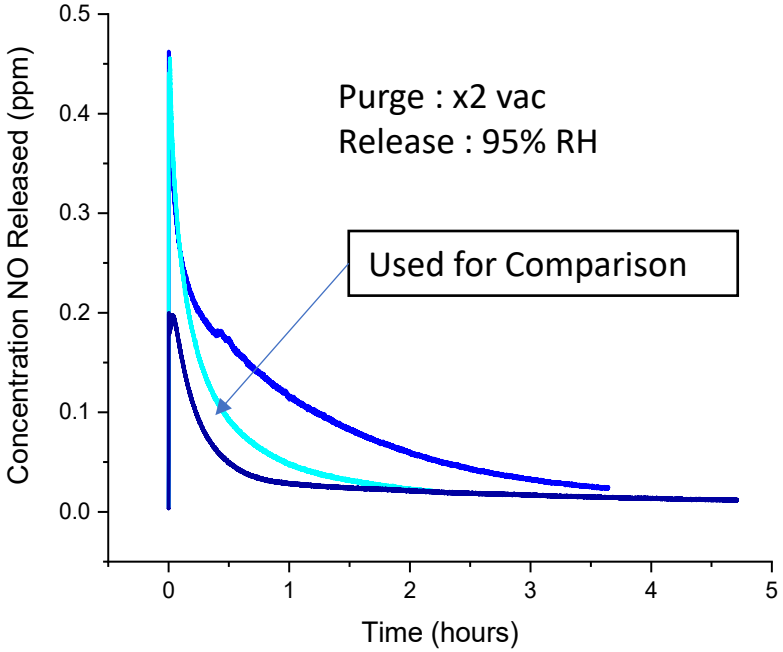
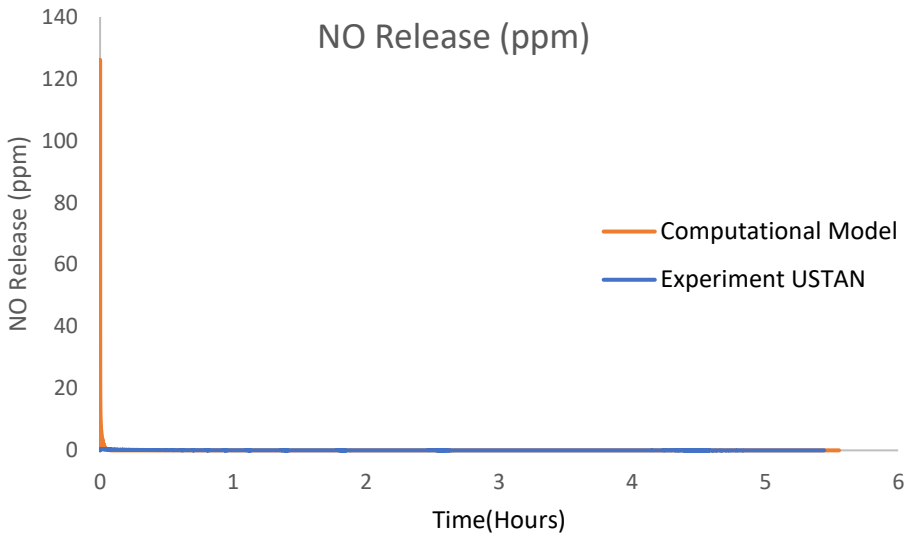
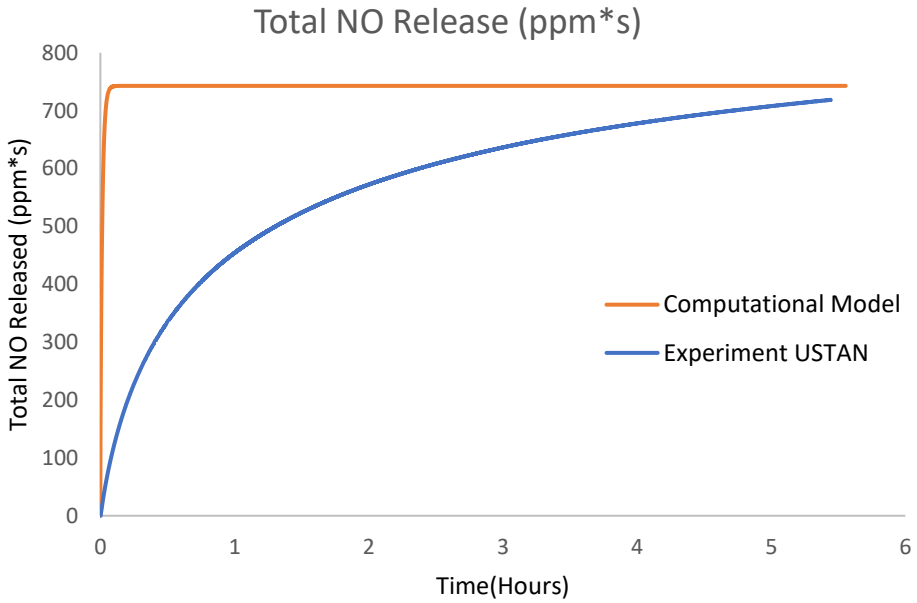






NO release for blank Pebax 5533 95% RH

Diffusivity of Material

NO Diffusivity : 1.2e-6 cm2/s
From Mecadi Tests

Solubility of Material

NO Solubility : Derived From Mecadi Results

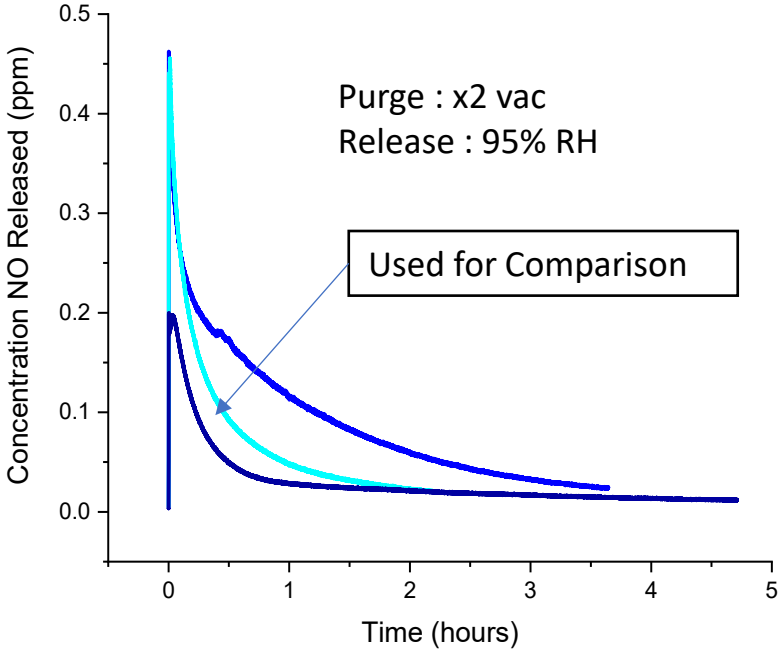
NO Loading / IC

Total NO release from Experiments is Input for the Model:

Release Conditions/BC

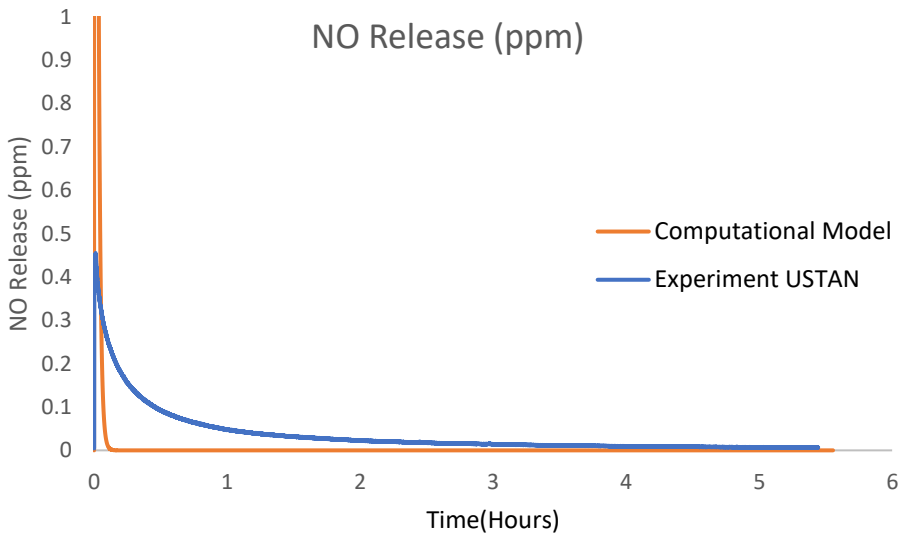
95% RH

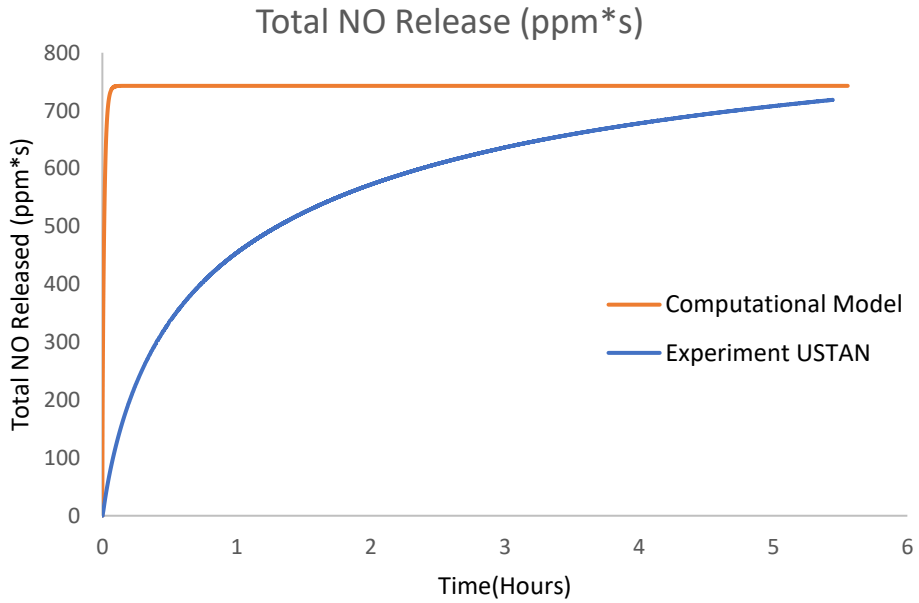
Dehydration: 80°C, 16 hr, x10^{-3} torr
NO Loading : 1 hr, 2 bar, 100%



NO release Test on 5533 Blank x2 vac

Compared to the Model The NO release in experiments shows extremely slow diffusion.





NO release for blank Pebax 5533 95% RH

What if , Diffusion coefficient is reduced by 100x?

1/100x

Diffusivity of Material

NO Diffusivity : 1.2e-8 cm2/s
From Mecadi Tests (1/100x)

Solubility of Material

NO Solubility : Derived From Mecadi Results

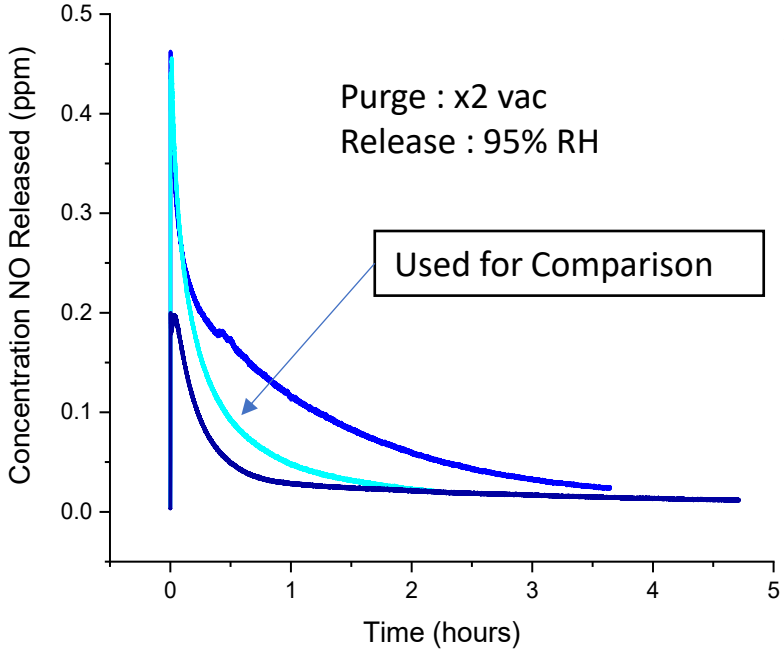
NO Loading / IC

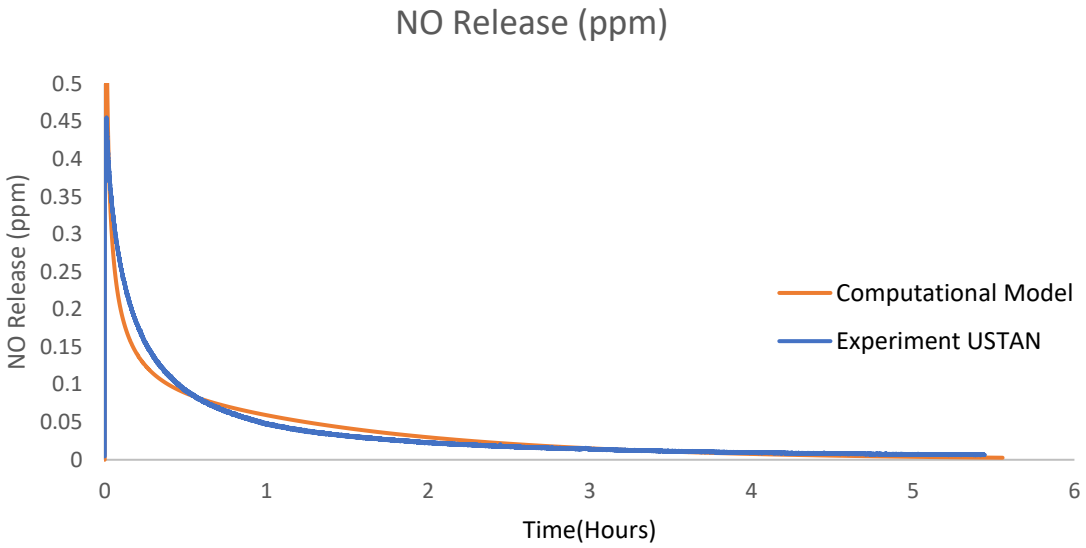
Total NO release from Experiments is Input for the Model:

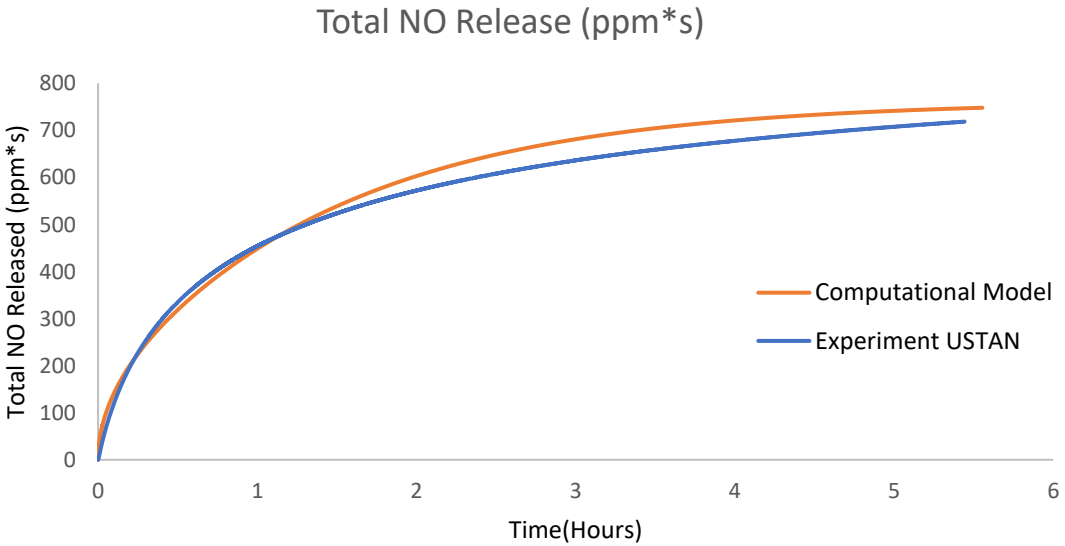
Release Conditions/BC

95% RH

Dehydration: 80°C, 16 hr, x10^{-3} torr
NO Loading : 1 hr, 2 bar, 100%







NO release for 5wt% Ni Pebax 5533 95% RH

Diffusivity of Material

NO Diffusivity : 1.2e-6 cm2/s
H2O Diffusivity : 2.2e-7 cm2/s
From Mecadi Tests

Solubility of Material

NO Solubility : Derived From Mecadi Results
H2O Solubility : Derived From Mecadi Results

NO Loading / IC

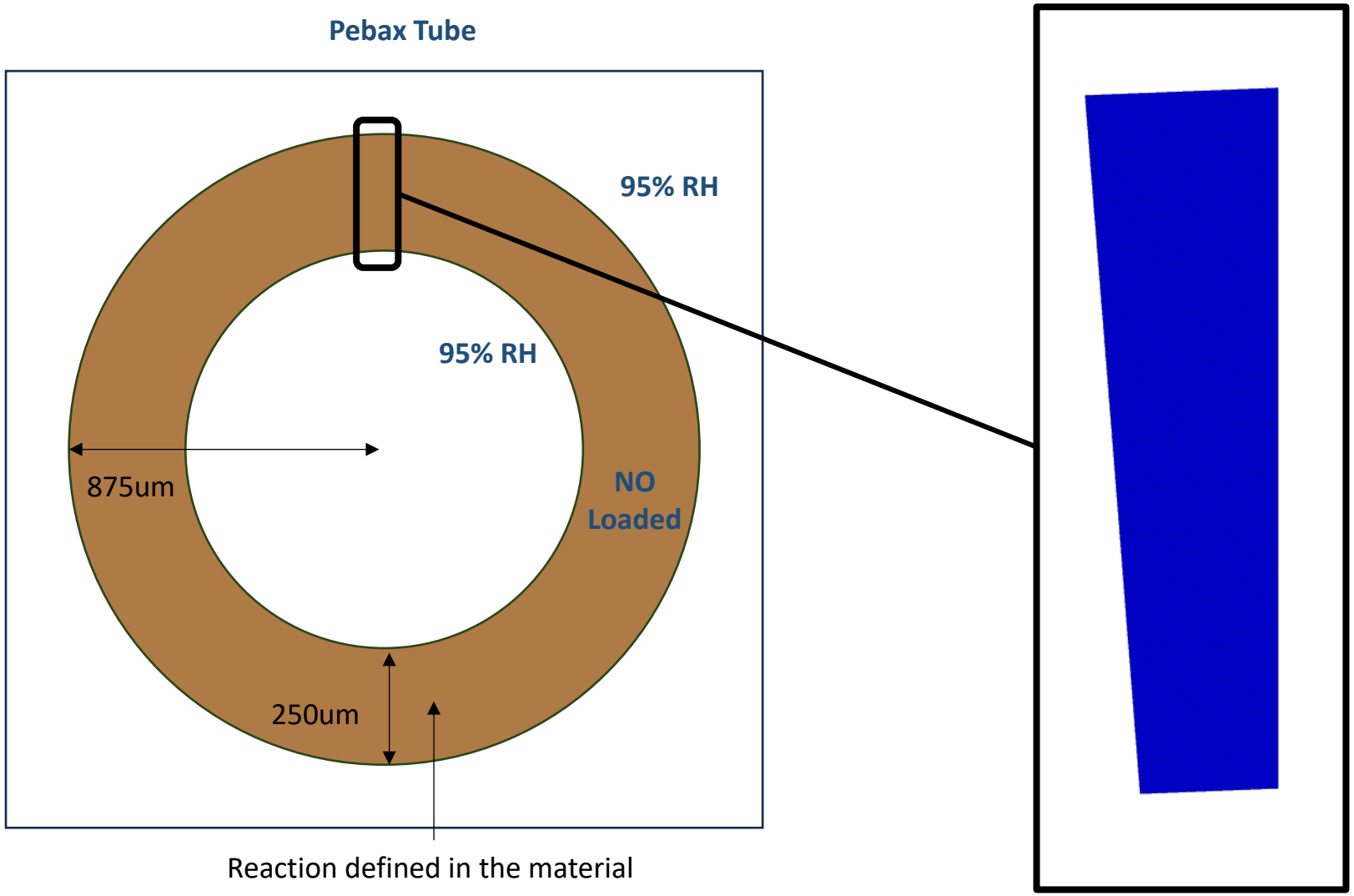
Total NO release from Experiments is Input for the Model:

Release Conditions/BC

95% RH

Rate of Reaction

Instantaneous



Reaction defined in the material

$$(\text{Material} + NO_I) + H2O \longrightarrow (\text{Material} + H2O) + NO_R$$

NO release for 5wt% Ni Pebax 5533 95% RH

Only Chemisorbed NO

Diffusivity of Material

NO Diffusivity : 1.2e-6 cm2/s
H2O Diffusivity : 2.2e-7 cm2/s
From Mecadi Tests

Solubility of Material

NO Solubility : Derived From Mecadi Results
H2O Solubility : Derived From Mecadi Results

NO Loading / IC

Total NO release from Experiments is Input for the Model:

Release Conditions/BC

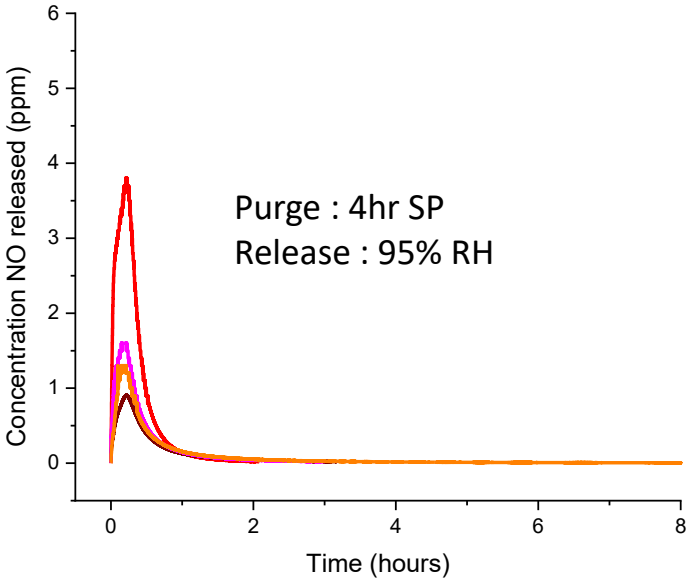
95% RH

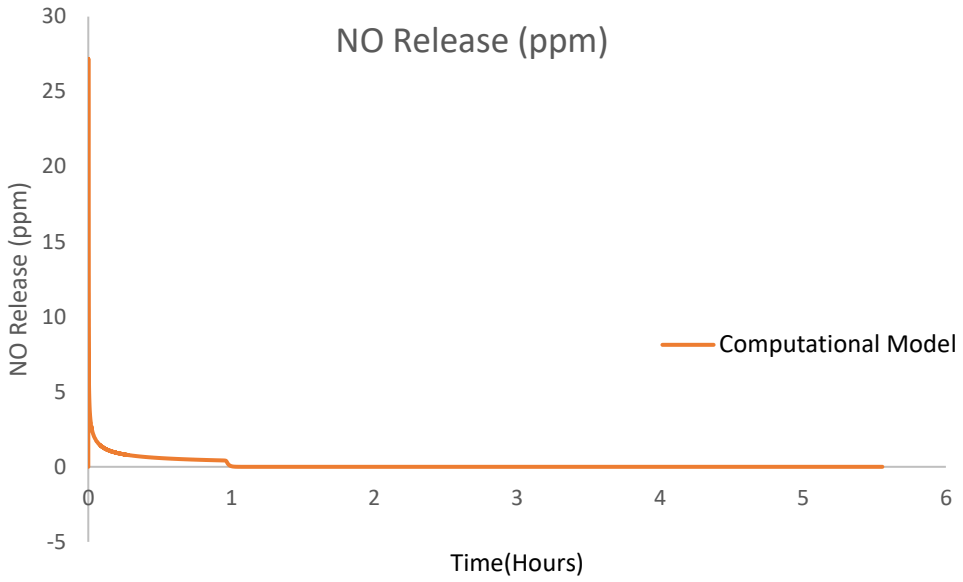
Rate of Reaction

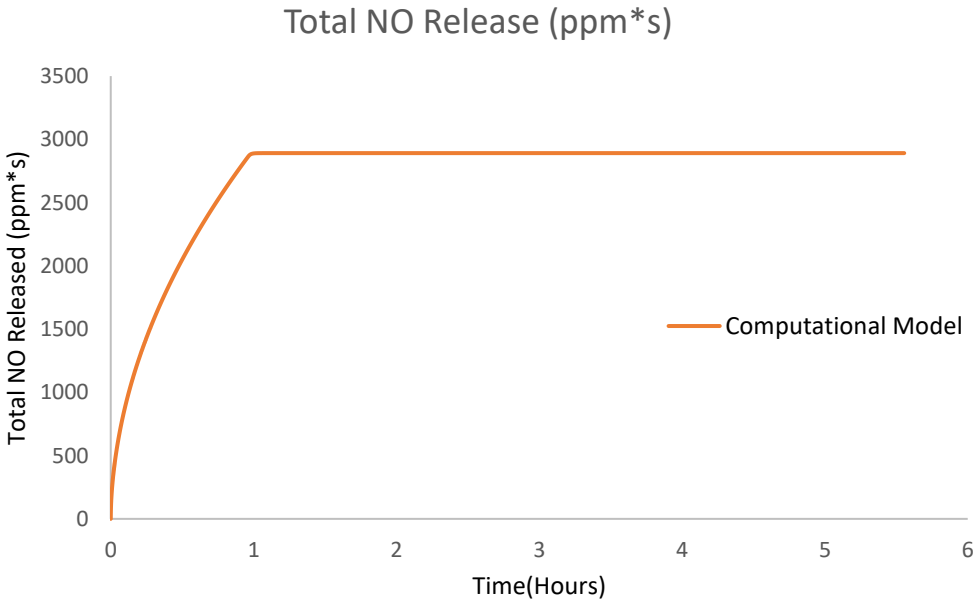
Instantaneous

Dehydration: 80°C, 16 hr, x10^{-3} torr
NO Loading : 1 hr, 2 bar, 100%

Model will represent the 80°C Dehydrated Sample results more Accurately







NO release for 5wt% Ni Pebax 5533 95% RH

Only Chemisorbed NO

Diffusivity of Material

NO Diffusivity : 1.2e-6 cm2/s
H2O Diffusivity : 2.2e-7 cm2/s
From Mecadi Tests

Solubility of Material

NO Solubility : Derived From Mecadi Results
H2O Solubility : Derived From Mecadi Results

NO Loading / IC

Total NO release from Experiments is Input for the Model:

Release Conditions/BC

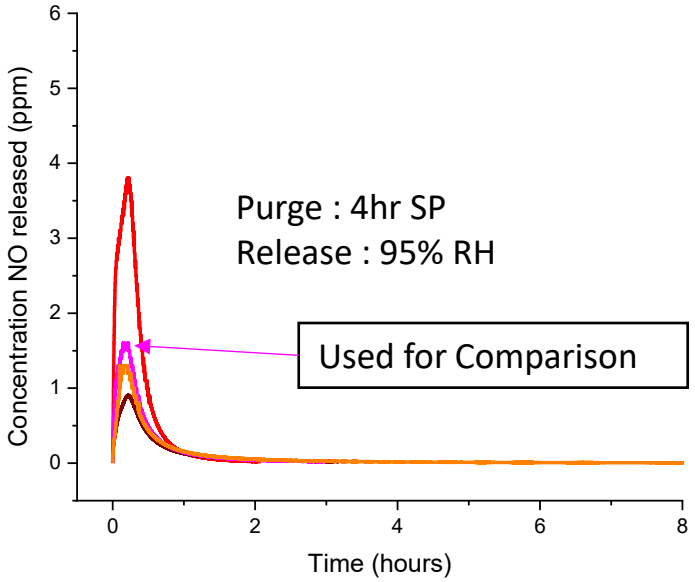
95% RH

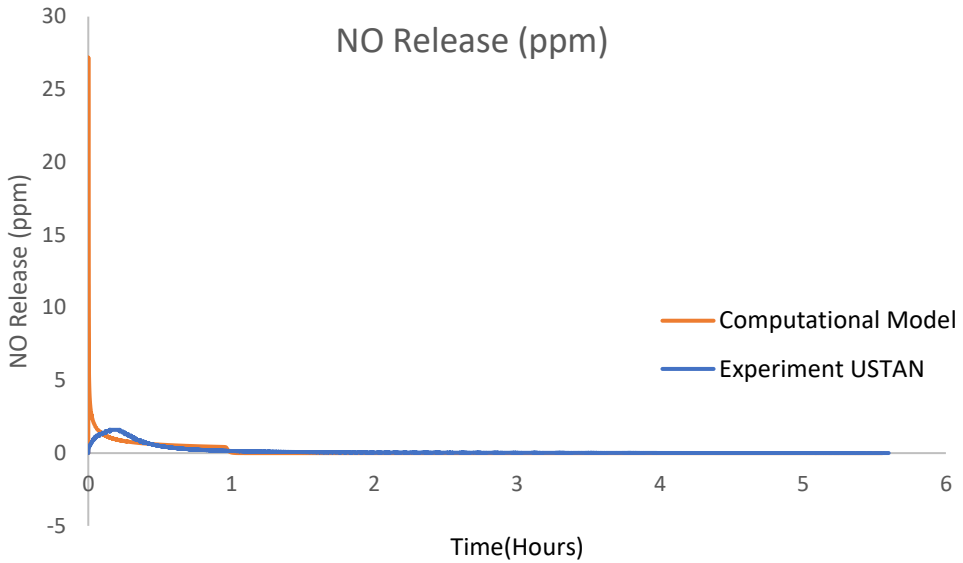
Rate of Reaction

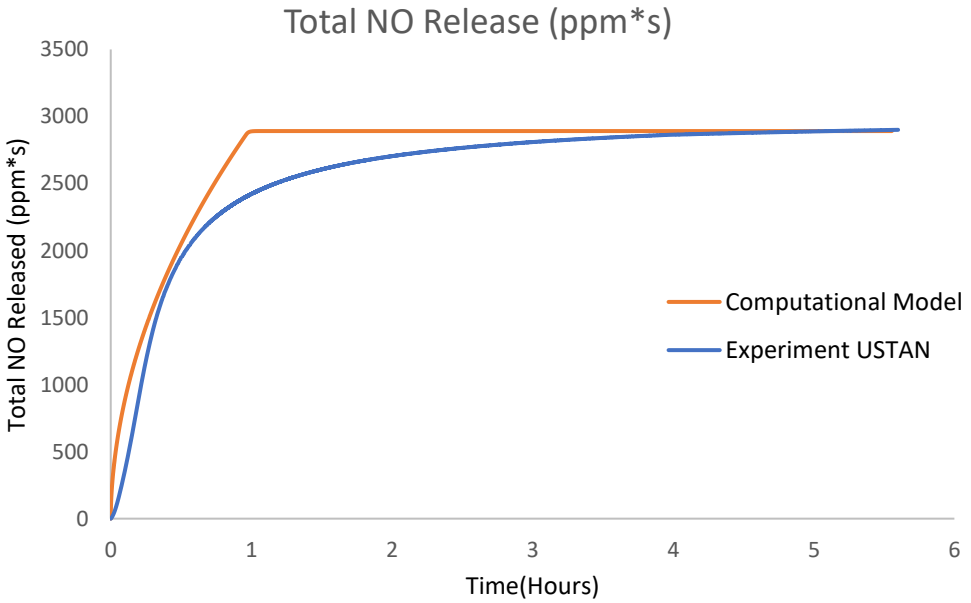
Instantaneous

Dehydration: 80°C, 16 hr, x10^{-3} torr
NO Loading : 1 hr, 2 bar, 100%

Model will represent the 80°C Dehydrated Sample results more Accurately







NO release for 5wt% Ni Pebax 5533 95% RH

Only Chemisorbed NO

Diffusivity of Material

NO Diffusivity : 1.2e-6 cm2/s
H2O Diffusivity : 2.2e-7 cm2/s
From Mecadi Tests

Solubility of Material

NO Solubility : Derived From Mecadi Results
H2O Solubility : Derived From Mecadi Results

NO Loading / IC

Total NO release from Experiments is Input for the Model:

Release Conditions/BC

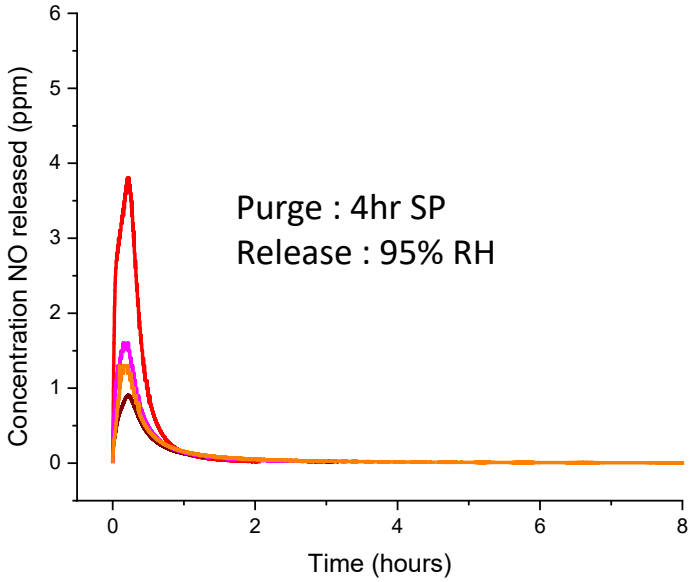
95% RH

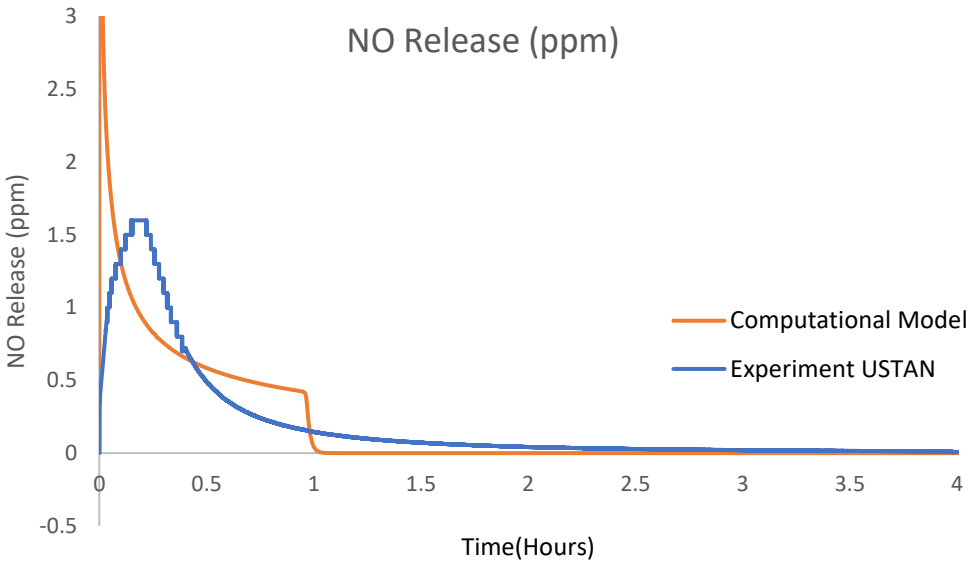
Rate of Reaction

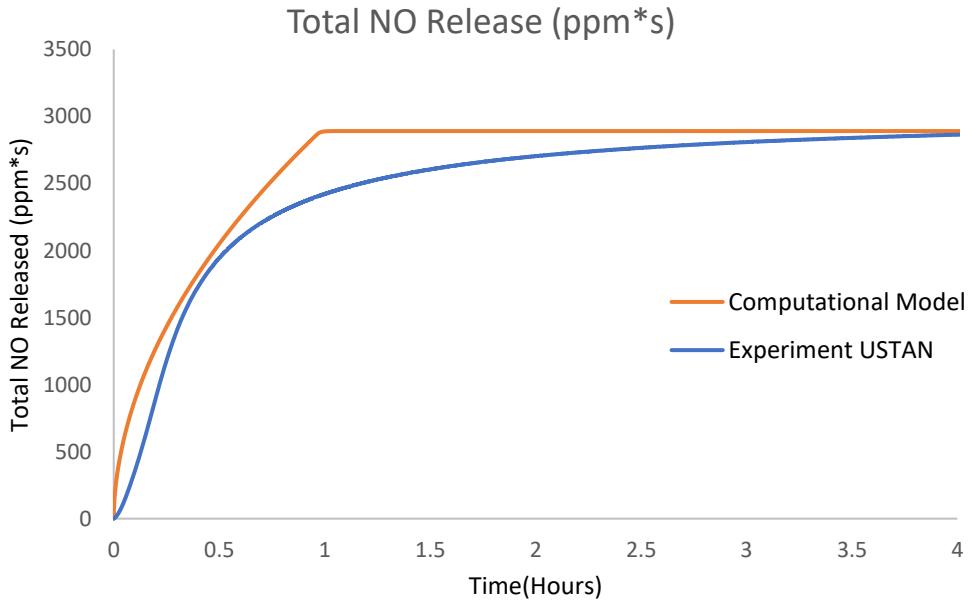
Instantaneous

Dehydration: 80°C, 16 hr, x10^{-3} torr
NO Loading : 1 hr, 2 bar, 100%

Model will represent the 80°C Dehydrated Sample results more Accurately



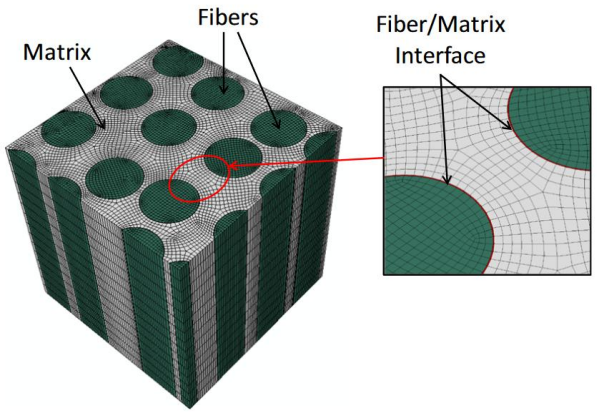




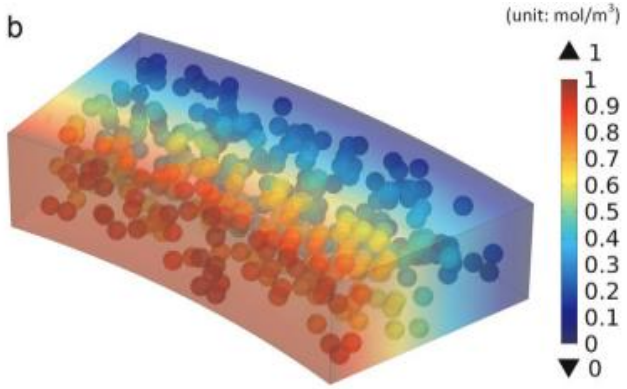
Discussion & Prospectives: Future Prospective

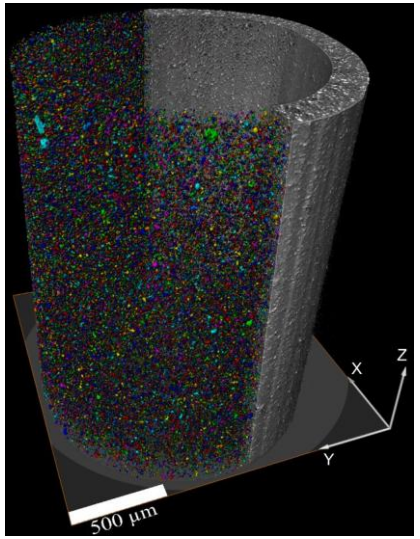
Continuum Model Development → Redefined Model Based on Experimental Data → Simulating Large Dataset → Updated Surrogate Model → Final result Verification and Model Validation

NO Diffusion Tests | NO Release Tests | H2O Diffusion Tests



RVE(Representative volume element) model





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Floor is open for Questions and Discussions.

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